

Experiment - 7.1.1. Inverted Star Pattern

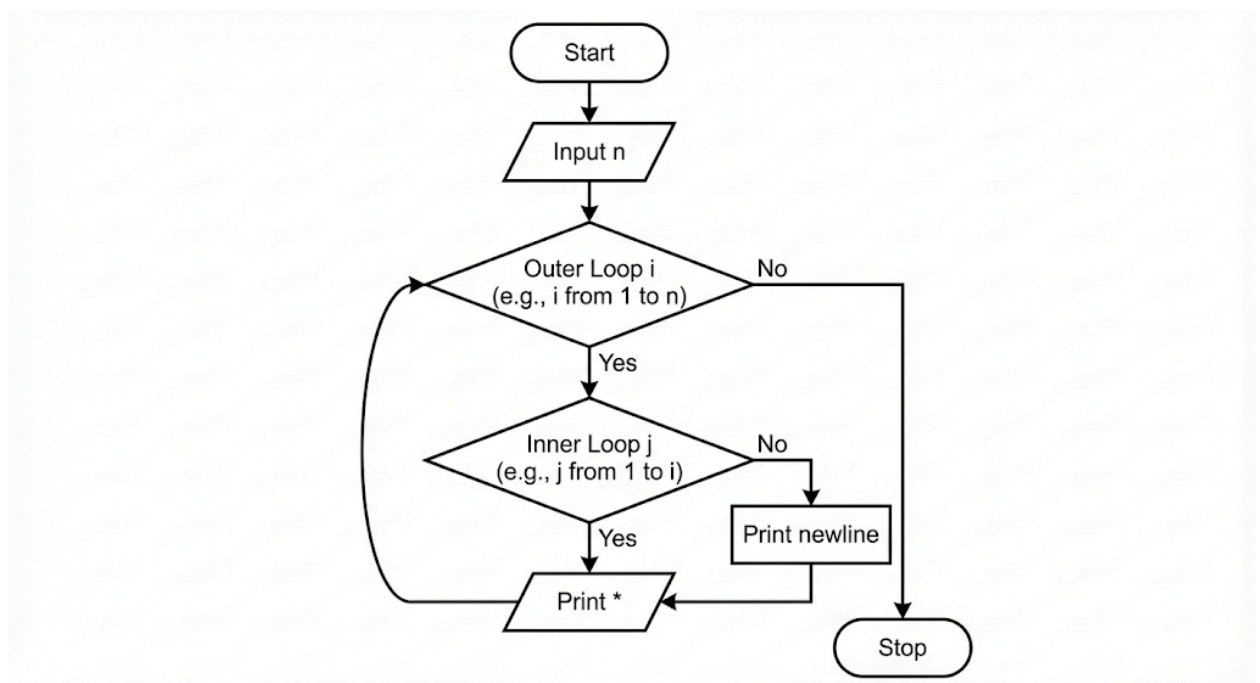
1. Aim

To design and implement a Python program that prints an inverted right-angled triangle star pattern. For a given integer n , the program uses nested loops to print n rows of stars, starting with n stars in the first row and decreasing by one star in each subsequent row until the last row has 1 star.

2. Pseudocode

1. **START**
2. **READ** a single integer from the user and **STORE** it in the variable n .
3. **FOR** each number i in the range starting from n down to 1 (decrementing by 1):
 - **FOR** each number j in the range from 0 to $i - 1$:
 - **PRINT** a star "*" without a newline character at the end.
 - **END INNER FOR**
 - **PRINT** an empty line (newline character) to move to the next row.
4. **END OUTER FOR**
5. **END**

3. Flowchart



4. Python Program

```
n = int(input())
```

```
for i in range(n, 0, -1):  
    for j in range(i):  
        print("*", end="")  
    print()
```

5. Experiment Screenshot

The screenshot displays the CodeTANTRA IDE interface. On the left, a panel titled "7.1.1. Inverted Star Pattern" provides instructions for writing a Python program to print an inverted right-angled triangle star pattern. It specifies that for a given number n , the program should print n rows of stars, with the first row containing n stars and each subsequent row having one less star than the previous row. It also notes that all rows should be left-aligned with no spaces between the stars. The input format is a single line containing an integer n , and the output format is an inverted right-angled triangle star pattern with n rows. A note refers to sample test cases for better understanding.

The main editor area shows the following Python code:

```
1 n = int(input())  
2  
3 for i in range(n, 0, -1):  
4     for j in range(i):  
5         print("*", end="")  
6     print()
```

Below the code editor, the execution results are displayed. The average time is 0.003 s and the maximum time is 0.003 s. The output shows that 3 out of 3 shown test case(s) passed and 3 out of 3 hidden test case(s) passed. A detailed view of Test case 1 shows the expected output as an inverted right-angled triangle star pattern with 3 rows.